

Activity 8: HackerRank Algorithmic Problem-Solving & Portfolio Integration

Portfolio Building (B25CS0311) | 3rd Semester B.Tech, Computer Science & Engineering

Student	Trivikram Kalagi
GitHub Repository	https://github.com/trivikramkalagi91-commits/HackerRank-3rdSem-Portfolio
HackerRank Profile	https://www.hackerrank.com/profile/trivikramkalagi1
Language / Badges	C++ Problem Solving 3-Star Badge and C++ 3-Star Badge earned

1. Problem Set and Complexity Analysis

#	Problem	Topic	Score	Status	Time	Space
1	Diagonal Difference	2D Arrays / Matrices	10 / 10	Accepted	O(N)	O(1)
2	Dynamic Array	Vectors / Sequences	15 / 15	Accepted	O(N + Q)	O(N + Q)*
3	Time Conversion	Strings & Logic	15 / 15	Accepted	O(1)	O(1)
4	Compare the Triplets	Basic Implementation	10 / 10	Accepted	O(1)	O(1)
5	Sparse Arrays	Hash Maps / Strings	25 / 25	Accepted	O(N + Q)	O(N)

* Dynamic Array stores up to Q appended values across N sequences plus the answer list, so auxiliary space is O(N + Q). N = input size (matrix dimension or number of strings), Q = number of queries.

2. HackerRank Evidence: Accepted Submissions and Badges

Prepare > Algorithms > Warmup > Diagonal Difference

Diagonal Difference ★

Problem

Submissions

Leaderboard

Discussions

Editorial

You made this submission 8 days ago.

Score: 10.00 Status: Accepted

People who solved **Diagonal Difference** attempted this next:

Birthday Cake Candles

Determine the number of candles that are blown out.

Submitted Code

Language: C++20

Diagonal Difference: Accepted, 10/10 (C++20)

Prepare > Data Structures > Arrays > Dynamic Array

Dynamic Array ★

Problem

Submissions

Leaderboard

Discussions

Editorial

You made this submission 8 days ago.

Score: 15.00 Status: Accepted

People who solved **Dynamic Array** attempted this next:

Left Rotation

Given an array and a number, d, perform d left rotations on the array.

Submitted Code

Language: C++20

Dynamic Array: Accepted, 15/15 (C++20)

Prepare > Algorithms > Warmup > Time Conversion

Time Conversion ★

Problem

Submissions

Leaderboard

Discussions

Editorial

You made this submission 8 days ago.

Score: 15.00 Status: Accepted

People who solved **Time Conversion** attempted this next:

Tower Breakers

Given N towers of height M and the rules for breaking them down, determine who will win the battle.

Submitted Code

Language: C++20

Time Conversion: Accepted, 15/15 (C++20)

Prepare > Algorithms > Warmup > Compare the Triplets

Compare the Triplets ★

Problem

Submissions

Leaderboard

Discussions

Editorial

You made this submission 8 days ago.

Score: 10.00 Status: Accepted

People who solved **Compare the Triplets** attempted this next:

Birthday Cake Candles

Determine the number of candles that are blown out.

Submitted Code

Language: C++

Compare the Triplets: Accepted, 10/10 (C++)

Prepare > Data Structures > Arrays > Sparse Arrays

Sparse Arrays ★

Problem

Submissions

Leaderboard

Discussions

Editorial

You made this submission 8 days ago.

Score: 25.00 Status: Accepted

People who solved **Sparse Arrays** attempted this next:

Array Manipulation

Perform m operations on an array and print the maximum of the values.

Submitted Code

Language: C++20

Sparse Arrays: Accepted, 25/25 (C++20)



Trivikram Kalagi 
@trivikramkalagi1

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My Badges



Problem Solving



C++

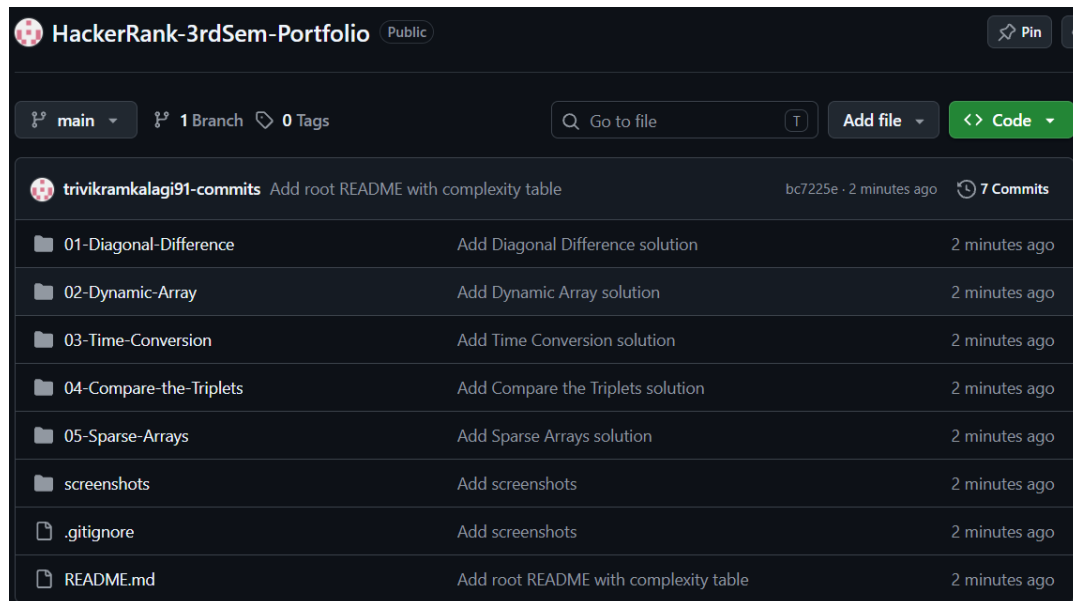
HackerRank profile: Problem Solving and C++ 3-Star badges

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3. GitHub Repository

The public repository <https://github.com/trivikramkalagi91-commits/HackerRank-3rdSem-Portfolio> contains one folder per problem (01-Diagonal-Difference to 05-Sparse-Arrays), each with a commented `solution.cpp` and a short README, a screenshots/ folder, and a root README with the profile link, evidence and complexity table. The work was committed in seven separate, descriptively named commits (one per solution, then screenshots, then the root README).



Repository home page showing the folder hierarchy and commit history

4. Reflective Summary: Algorithmic Optimization Techniques

Working through these five problems showed me that optimization usually comes from choosing the right data structure rather than writing clever code. In Diagonal Difference, one loop using indices i and $n-1-i$ sums both diagonals together, giving $O(N)$ time and $O(1)$ space instead of scanning the whole matrix. In Sparse Arrays, comparing every query against every string would cost $O(N \times Q)$; building an `unordered_map` of frequencies once makes each lookup $O(1)$ on average and the total $O(N + Q)$. Dynamic Array taught me to treat a vector of vectors as an indexable structure, using XOR with `lastAnswer` to select a sequence in constant time per query. Time Conversion and Compare the Triplets are constant-time problems, but they stressed edge cases such as 12:00:00AM becoming 00:00:00 and ties awarding no points. I also used fast input with `sync_with_stdio(false)` and passed containers by const reference to avoid copies. Documenting complexity beside each solution and committing them one by one made me justify every design choice. Earning the Problem Solving and C++ 3-Star badges confirmed that steady practice of fundamentals pays off.